

Cold & Wet Weather Installation Guidelines

Field requirements for bonding DeckRite vinyl membrane in cold, damp or changing weather conditions.

WHY THIS MATTERS

Cold or wet conditions can reduce adhesive transfer, slow cure time and increase the risk of condensation below the membrane. Confirm the substrate, humidity and dew point before installation and maintain proper conditions until adhesive cure.

Minimum installation conditions



AMBIENT AIR

40°F / 5°C
and rising



DECK SURFACE

40°F / 5°C
and rising



WOOD MOISTURE

Below 20%



SURFACE

Clean, dry,
frost-free

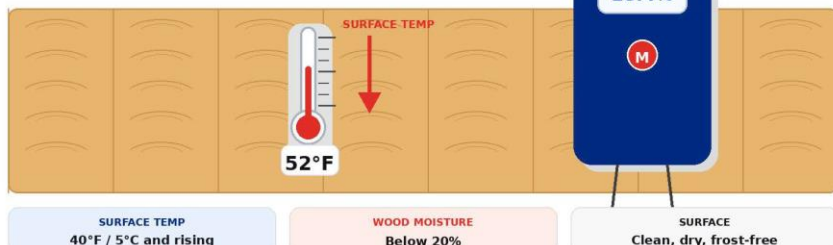


DEW POINT

Surface 5°F /
3°C above
dew point

Measure the surface — not just the air.

Cold weather failures often begin when a deck looks dry, but the surface is too cool, too damp, or near the dew point.



Dew point control

Dew point is the temperature where airborne moisture condenses on a surface. If the deck surface approaches the dew point, invisible moisture can form on the substrate or adhesive and interfere with bond strength.

FIELD RULE

Install only when the deck surface remains at least 5°F / 3°C above dew point.

DEW POINT CALCULATION CHART (FAHRENHEIT)

AT 30 Hg BAROMETRIC PRESSURE

Ambient Air Temperature °F

% Relative Humidity

	20	30	40	50	60	70	80	90	100	110
90	18	28	37	47	57	67	77	87	97	107
85	17	26	36	45	55	65	75	84	95	104
80	16	25	34	44	54	63	73	82	93	102
75	15	24	33	42	52	62	71	80	91	100
70	13	22	31	40	50	60	68	78	88	96
65	12	20	29	38	47	57	66	76	85	93
60	11	19	27	36	45	55	64	73	83	92
55	9	17	25	34	43	53	61	70	80	89
50	6	15	23	31	40	50	59	67	77	86
45	4	13	21	29	37	47	56	64	73	82
40	1	11	18	26	35	43	52	61	69	78
35	-2	8	16	23	31	40	48	57	65	74

Field Rule: Do not install unless the deck surface temperature stays at least 5°F (3°C) above the dew point during application and adhesive cure.

Example: At 70°F air temperature and 70% RH, means dew point is 60°F. Using the field rule of adding 5°F, surface temperature must be at least 65°F

Control the work environment

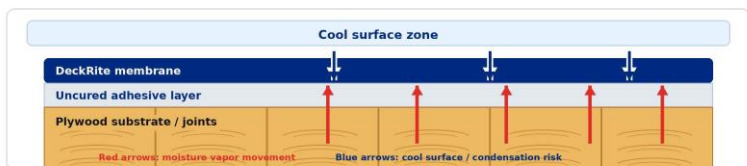
Prevent vapor drive, protect adhesive cure and avoid creating separate temperature zones above and below the deck.

DO NOT INSTALL

Do not install over wet plywood, frost, standing water or when temperatures are expected to fall below the dew point or freezing before the adhesive has cured.

Vapor Drive Risk

Warm, moist air can move toward cooler areas and condense beneath uncured membrane adhesive.



Warm, humid air below deck

Uneven heating or partial tarping can push moisture upward through plywood joints before adhesive has cured.

Result

Moisture can collect below the membrane and weaken the adhesive bond before cure is complete.

Vapor drive

Warm, humid air naturally moves toward cooler areas. If only one side of the deck is heated or tarped, moisture vapor can be driven through plywood joints and condense under uncured adhesive.

Temporary enclosure

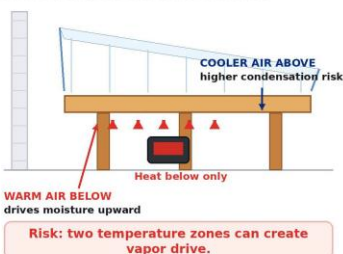
When weather is below acceptable limits, enclose the deck to the ground and heat the entire work area. The goal is one warm, dry environment around the substrate, membrane and adhesive.

Temporary Enclosures

Create one consistent warm, dry environment above and below the deck.

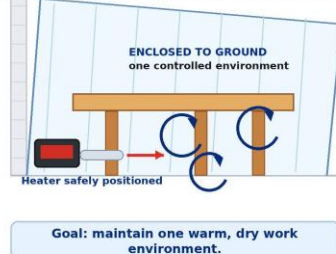
AVOID

Do not heat only one side of the structure.



PREFERRED

Enclose to the ground and heat the entire work area.



Adhesive handling guidance

ADHESIVE / CONDITION	WHAT TO CONTROL	FIELD ACTION
Solvent-based contact adhesive	Keep adhesive warm and protected from moisture. Solvents can cool the surface as they flash off.	If blushing occurs, allow moisture to evaporate and recoat only after proper conditions are restored.
Water-based adhesive	Avoid high humidity, rain and temperatures that may drop below dew point before cure.	Allow proper open time and curing conditions; do not trap wet adhesive under membrane.
All adhesives	Surface must be dry, clean and within temperature / moisture limits before application.	Follow current DeckRite adhesive instructions, labels and SDS requirements.

Installer checklist

- Confirm surface temperature is 40°F / 5°C and rising.
- Verify wood moisture content is below 20%.
- Check relative humidity and dew point.
- Maintain surface 5°F / 3°C above dew point.
- Protect installation until adhesive has fully cured.

TECHNICAL REMINDER

Weather cannot be controlled, but jobsite conditions can. Monitoring temperature, moisture, humidity and dew point greatly improves the likelihood of a successful membrane installation.

Examples of Vapor Drive Under Vinyl Before Adhesive/Joint Compound Has Cured

Field photos showing visible ridges, wrinkles or telegraphed lines that can appear when moisture pressure develops below uncured adhesive and/or joint compound.



Note: Vapor-drive appearance can vary by membrane color, substrate condition, adhesive type, temperature change and jobsite humidity.